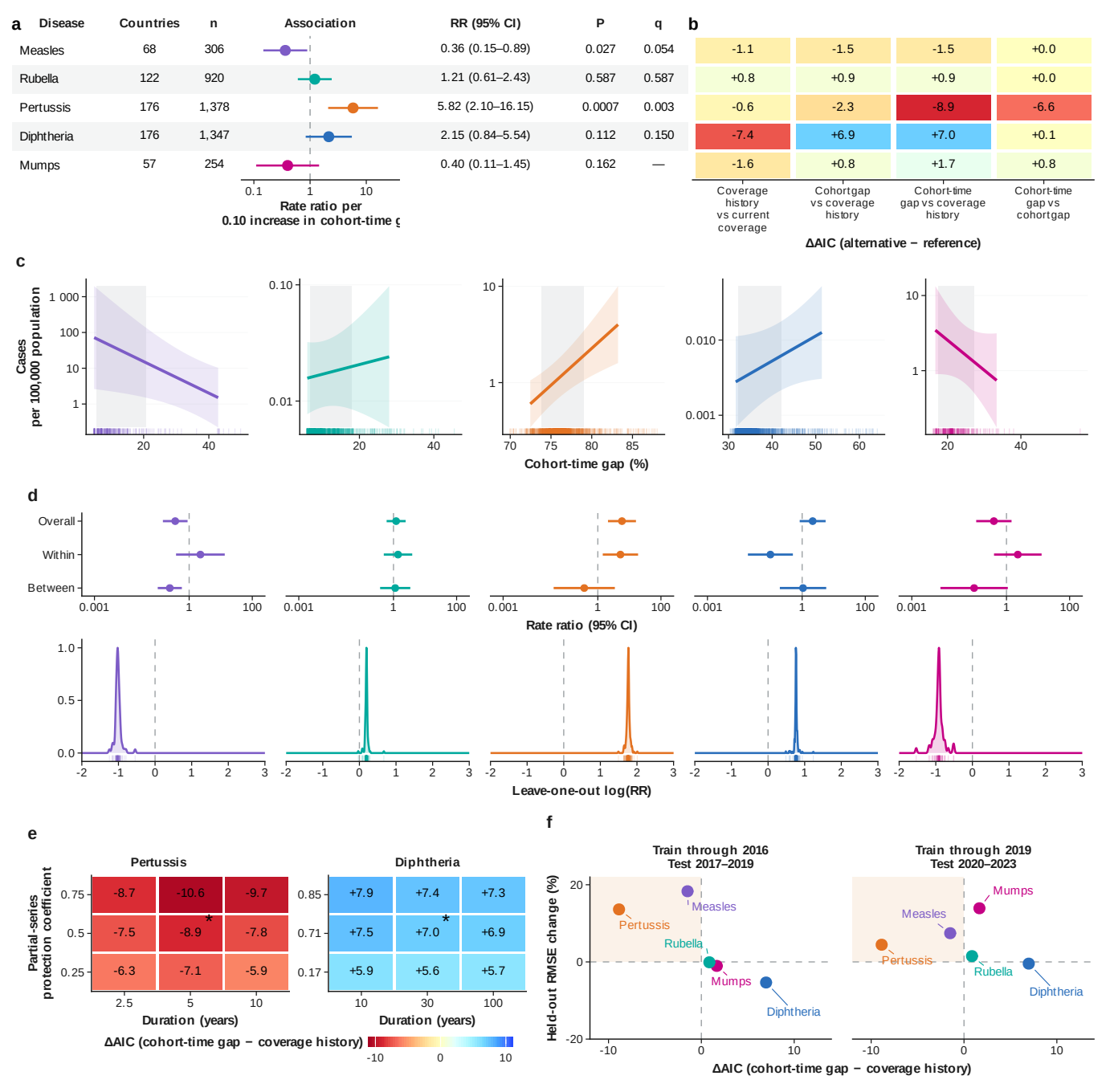

a
Disease
Countries
n
Association
RR (95% CI)
P
q
Measles
68
306
0.36 (0.15–0.89)
0.027
0.054
Rubella
122
920
1.21 (0.61–2.43)
0.587
0.587
Pertussis
176
1,378
5.82 (2.10–16.15)
0.0007
0.003
Diphtheria
176
1,347
2.15 (0.84–5.54)
0.112
0.150
Mumps
57
254
0.40 (0.11–1.45)
0.162
—
0.1
1
10
Rate ratio per
0.10 increase in cohort-time g
b
-1.1
-1.5
-1.5
+0.0
+0.8
+0.9
+0.9
+0.0
-0.6
-2.3
-8.9
-6.6
-7.4
+6.9
+7.0
+0.1
-1.6
+0.8
+1.7
+0.8
Coverage history vs current coverage
Cohortgap vs coverage history
Cohort-time gap vs coverage history
Cohort-time gap vs cohortgap
ΔAIC (alternative − reference)
c
Cases per 100,000 population
1 000
100
10
1
0.10
0.01
10
1
0.010
0.001
20
40
70
75
80
85
30
50
60
Cohort-time gap (%)
d
Overall
Within
Between
0.001
1
100
Rate ratio (95% CI)
1.0
0.5
0.0
-2
-1
0
1
2
3
Leave-one-out log(RR)
e
Pertussis
Diphtheria
Partial-series protection coefficient
0.75
0.5
0.25
0.85
0.71
0.17
-8.7
-10.6
-9.7
-7.5
-8.9
-7.8
-6.3
-7.1
-5.9
+7.9
+7.4
+7.3
+7.5
+7.0
+6.9
+5.9
+5.6
+5.7
*
2.5
5
10
30
100
Duration (years)
ΔAIC (cohort-time gap − coverage history)
-10
0
10
f
Train through 2016
Test 2017–2019
Train through 2019
Test 2020–2023
Held-out RMSE change (%)
20
0
-20
Measles
Pertussis
Rubella
Mumps
Diphtheria
-10
0
10
ΔAIC (cohort-time gap − coverage history)